Action, Trajectories and Gaps: A Research Programme

navstokgap research working notes

9 September 2026 — programme version 22

Abstract

We organize the study of mechanical action scales around exact trajectories, fluctuation spectra and operational resolution. A sequence from constant force through oscillator and central-potential models leads to two research targets: quantum reconstruction and a mechanism for gap formation. Each work package has a mathematical object, deliverable and acceptance gate. Relativistic and PDE/field-theory comparisons track how estimates behave under changes of scale. The maintained source is `research/PROGRAMME.md`; task state and handoffs support continuation across research sessions.

1 Research programme: action, trajectories and gaps

Version 48, 2026-09-12.

1.1 Aim

Hard verification rule (2026-09-09): mathematical work uses written proofs and source review. Creating or running Python numerical/symbolic verification scripts is prohibited for coordinator and workers. Existing scripts are historical artifacts, even where older task entries mention checks. Document and source-management tooling remains allowed; switching languages is not a workaround. A new computational-verification workflow requires user direction.

Explain the relationship between mechanical trajectories, an action scale and gap formation. We pursue two connected outcomes:

1. A reconstruction of quantum mechanics from explicit physical premises, with a derivation of the role and necessity of an action parameter.
2. A rigorous toy model that identifies which Lagrangians, state spaces and boundary conditions produce a positive lower bound.

Newton's shrinking-area construction provides the initial geometric model. The work proceeds through exact mechanical examples, operational definitions, operator spectra and comparisons of limiting procedures.

The user's strong target (2026-09-06) is to identify independently motivated physical principles that force quantum structure with a positive action parameter, with classical mechanics recovered as a specified limit. This has two proof obligations: exclusion of the classical model under the selected principles, and construction of the classical limit. The statement that every admissible classical theory has a quantum completion is a further universal claim whose domain and notion of quantisation must be fixed.

Solvable-model spectra and sector gaps are test cases for this programme. Each result has a separate librarian audit of prior literature, recording established matches, derived consequences or a candidate contribution with explicit search coverage. The strong target receives its own prior-art audit as its axioms and quantifiers become precise.

1.2 Current priority: classical cuts and receiver memory

R03/B33 are complete: the cut-state note proves that stationary position-conditioned kernels in a fixed-energy spring receiver fail composition. Independent direction resets freeze terminal position under refinement while leaving one-time marginals invariant. Full phase state restores exact composition, and its action E/ω remains prepared. C062–C063 have a two-page Luna-low audit and coordinator written/source review. **R04/B34 are complete:** the three-body note derives the surviving receiver memory with tagged momentum retained, two-time phase reconstruction and its short-separation conditioning. C064–C065 separate the fixed coupling kernel from prepared action scales that close under cooling. **R05/B35 are complete:** the readout note proves that shrinking classical probe widths can overcome cubic observation sensitivity with vanishing accumulated disturbance. C066–C067 record the conditional result for ideal impulses, exact pointer records and a growing probe supply. **R06/B37 are complete:** a smooth autonomous apparatus with one clock and four probes has fixed finite mass, fixed pulse widths and uniform force ceilings. Weak coupling and concentrated preparations give accurate, weakly disturbing delayed reconstruction; its specified action product tends to zero (C068–C069). Exact final records and known dynamics are supplied. **R07/B38 are complete:** exact delayed records plus unresolved bounded force give sharp prediction-risk lower bounds and a joint reachable canonical area $2F^2\ell^3/(3m)$ (C070–C071). The input class and delay supply this scale; it closes when either shrinks. **R08/B39 are complete:** exact joint reachable sets compose across unobserved cuts; an observed position leaves a calculable momentum fibre and terminal area (C072–C073). That area can close while momentum uncertainty persists. **R09/B40 are complete:** finite-precision records give an exact clipped-lens area and uniform closure in the joint precision/delay limit (C074–C075). **R10/B41 are complete:** two positions recover momentum as well, with estimator-specific errors and an explicit vanishing action product (C076–C077); the differentiation bound has an exact prior-art match. **R11/B42 are complete:** a symmetric-fibre theorem and an established prepared trajectory pair give exact phase minimax risks, even with the entire noisy position history (C078–C079). **R12/B43 are complete:** product minimax radii add linearly, so identical copies have an extensive canonical risk product. Aggregate records have an exact single-body image and strictly larger momentum risk when force/precision ratios differ (C080–C081). Mass-only coordinate-radius closure forces an extensive product; A08's variance universality does not transfer. **R13/B44 are complete:** the exact finite-horizon phase risks have three momentum regimes and earlier position saturation. Mixed constituent regimes can lose both phase coordinates under record aggregation (C082–C083). **R14/B45 are complete:** shared l^r record budgets give exact centre risks and saturated copy-count exponent 1-3/(2r), with invariance at r=3/2 (C084–C085). **R15/B46 are complete:** independent identical apparatus blocks restore an extensive saturated product even at r=3/2. Exact finite-horizon block risks and resource-set comparisons give C086–C087. **R16/B47 complete:** exact nonlinear calibration closes reconstruction/action products at fixed nonzero coupling when incoming preparation and final-record errors contract (C088–C089). **R17/B48 complete:** unknown incoming probe momenta preserve one exact record across a receiver-shell patch, giving positive preparation-dependent reconstruction bounds (C090–C091). **R18/B49 complete:** revealing incoming momenta leaves exact position-mediated ambiguity in a fixed pulse design: clock reaction gives an invertible response and a shell patch of radius proportional to coupling times preparation width (C092–C093). **R19/B50 complete:** eight final pointer coordinates uniformly recover receiver state and unknown incoming positions on a fixed preparation box, with known initial clock data and probe momenta. Improving final record precision closes the canonical reconstruction product (C094–C095). **R20/B51 complete:** hidden clock offset and speed leave an exact common-record shell curve in a fixed pulse design, giving positive preparation-dependent canonical risks (C096–C097). **R21/B52 complete:** exact receiver energy and revealed clock position give uniform local recovery even with clock momentum unknown; the canonical error product closes with final record errors on a fixed prepared patch (C098–C099). **R22/B53 complete:** two distinct clock speeds give exact equal-energy, equal-record receiver pairs across the full shell, persisting at positive coupling with canonical risk bounds (C100–C101). **R23/B54 complete:** final clock momentum restores uniform global recovery on a bounded receiver domain, without exact energy or local patch information; record-error products close with preparation fixed (C102–C103). **R24/B55 complete:** full final clock phase removes initial clock calibration while preserving global recovery (C104–C105). The R19–R24 comparison isolates known incoming probe momenta as the shared preparation premise. **R25/B56 complete:** a full unknown apparatus box permits exact compensation of all ten final records with interior margins (C106). At fixed box width and sufficiently weak coupling, the entire receiver shell has one record and coordinate minimax risks equal their no-record values (C107). **R26/B57 complete:** exact initial apparatus energy still permits an antipodal receiver pair with identical complete records; a backward terminal construction preserves both energies and preparation margins (C108–C109). **R27/B58 complete:** a known nonzero initial probe displacement still admits exact equal-record receiver pairs. Borsuk–Ulam on a constrained preparation sphere and the flow estimate give a positive full-state risk (C110–C111). **R28/B59 complete:** a fixed smooth pulse design gives an exact common-record curve separated in both x and P under the same calibration and energies. The canonical risk product stays positive at weak coupling with fixed preparation (C112–C113). **R29/B60 complete:** a second calibrated displacement leaves two exact common-record branches with local inverses at each and a positive global canonical risk bound (C114–C115). **R30/B61 complete:** three calibrations give uniform global recovery on a sufficiently small fixed apparatus box without exact receiver energy (C116–C117). The canonical precision bound scales as (record error/coupling) squared. **R31/B62 complete:** finite calibration tolerance gives receiver error O((epsilon+delta)/lambda) and, for exact final records, optimal canonical product of order min(1,(epsilon/lambda)^2). One compatible ball fills the receiver domain at sufficiently weak coupling, attaining exact no-record risks (C118–C119). **R32 stage 1/B63:** a single uncertain constraint leaves an exact receiver curve with leading direction L^{-1}e_j. Both canonical components being nonzero gives a positive error product on a zero-area curve (C120). Next evaluate the actual pulse-specific components, starting with apparatus energy j=4. This continues I005; A19 supports the main track.

The user's discovery-by-connection strategy is saved in I006. Each proposed bridge must specify its source and destination observables, preserved premises and a decisive test. R11 joins reachability to optimal recovery; R12 resolves the connection to A08 by deriving the different worst-case aggregation law. R13 resolves its finite-horizon boundary; R14 shows how error geometry selects the exponent. R15 settles the regrouping test: fresh independent block budgets multiply the product by the block count. R16 removes a systematic calibration remainder; R17 proves a fixed-preparation ambiguity; R18 locates its position-mediated successor. R19 removes it with full pointer records and known initial clock data. R20 tests the cost of that clock information.

1.3 Supporting source route: static compatibility

H10 supplies a question before the time cut: what allows distinct local contacts or observations to belong to one whole? The selected Indian, Chinese and Avicennian passages motivate explicit interfaces, controlled spatial remainders and apparatus relations. I007 proposes K01: specify fixed-time measurement arrangements and their overlaps, test classical joint-state models, then identify where an action-valued calibration enters. R12's exact cost of record aggregation is a benchmark; R32 keeps main-track priority. H11 strengthens the Chinese facsimile and Arabic translation witnesses.

1.4 Supporting track: bound action and scale selection

M07/B28 are complete. The Kepler note and gap laboratory specify the regular bound domain and angular-action threshold k/c (C052). A softened core admits bound circles for every positive angular action at fixed coupling and speed limit (C053); the core-collapse limit explains the domain change. One Luna-low follow-up audit and coordinator written proof/source review support these results.

A15/B29 are complete: the bound-orbit note compares transport and conditional variance with twice the canonical covariance area. The latter equals orbital action for uniform circular phase; each phase preparation conserves its own value. C054–C055 include the window-ratio bound and singular/softened threshold transfer. **A16/B30 are complete:** the dilation note proves a conditional zero-infimum theorem for contraction-closed model classes (C056–C057). **A17/B31 are complete:** one fixed smooth force-bounded potential has stable small circles of vanishing action, while an independent speed floor gives a positive circular bound (C058–C059). **A18/B32 are complete:** total turning extends this sharp bound to closed regular trajectories (C060–C061), with Milnor's curvature theorem audited. **A19 is next:** test a peak-excitation premise allowing speed to vanish at turning points. The singular-core benchmark remains coupling-dependent; the softened model tests the extra premise needed for a positive infimum. B27 supplies the established Boyer classification and B28 the bounded follow-up.

Keep the transport laboratory and its preparation tests. **A14** is the next bounded renewal calculation; **A16** tests scale-changing transformations against the axioms. **R02** audits the repaired finite-propagation draft, and **G03** extends spectral control beyond independent products. The new idea card specifies the observables and limits; TASKS governs dispatch. Quantum reconstruction Q01 and the Newton-specific historical source questions retain their separate gates.

1.5 Milestone history: composition and scale selection

The following dated milestones preserve the development of the programme; their next-step wording records earlier decisions, superseded by the priority above.

A13/B26 are complete. Ordered streams give zero response at the same mass, speed, density and mean collision rate as the Poisson receiver, through periodic cancellation. **A14** now tests stationary renewal streams at fixed mean gap: derive the tagged variance growth and how it depends on gap variance. The target is quantitative control of preparation, beyond a mean collision rate.

A12/B25 are complete. The collision manuscript derives the clock $\lambda = \rho u$ from independent spatial gaps in a two-speed equal-mass gas, giving $H_* = mu/\rho$ with a common hard speed ceiling. **A13** next changes spacing correlations at fixed mass, speed and density, testing which part of the response is selected by preparation.

A11/B24 are complete. The consolidated receiver paper gives a periodic chain plateau $\Theta\sqrt{m/k}$ in one order of limits, zero in the reverse order, and closure under a common hard speed ceiling in the stated phase preparation. **A12** next selects a bounded-velocity interacting receiver and its invariant preparation, before attempting a low-frequency response proof. A06/A09a are consolidated in one return-bridge manuscript; the paper review identifies further A01–A09 groupings while preserving distinct hypotheses.

A10/B23 are complete. Finite conservative harmonic receivers give an exact cosine covariance and vanishing long-window action observable at fixed centre velocity. Random centre velocity gives ballistic variance instead. **A11** now specifies an increasing network and its preparation: determine the tagged low-frequency spectral weight and compare the receiver-size and window limits. Harmonic-bath sources supply a reduced memory equation; its friction kernel must be distinguished from the tracer covariance. This advances the mechanical origin and preparation-independence gates of the action-selection target.

G02/B22 are complete. Weakening observable access can keep an action plateau fixed while the full gap closes, even with distinct velocity states. Independent composition instead lets local-frame bounds control mixed modes. **A10 is next:** a specified finite conservative harmonic receiver, with its normal-mode correlation, preparation dependence and centre-motion convention. Compare refinement, long observation and reservoir-size limits to determine which physical premise could sustain the positive action target. A09 retains the universal coherent-action identification.

A07/B21 are complete. The bounded-turn paper gives a positive endpoint-conditioned kinetic cost and a vanishing polygon error under uniform acceleration control. **G02 is next:** access to internal modes and observability under composition. The mechanical follow-up must specify a conservative receiver; A09 retains the universal scale-selection task.

G01/B20 are complete. The spectral-control note gives a lower gap bound from an observable frame and bounded summed susceptibility. Its hidden-label example keeps the velocity plateau fixed while the full gap closes. **A07 is next:** bounded-acceleration mechanical turns and their action scales. G02 tests observability under weaker access and composition. A09 retains the preparation-independent quantum-role target.

A09b/B18 are complete. The checkerboard note proves the normalized Dirac wavepacket limit and a repeated-measurement ballistic cut limit. C039–C040 separate a classical flip probability from a coherent corner amplitude. G01 supplies the subsequent spectral-control test. A09 retains the physical premise identifying the positive plateau with the coherent action parameter, across preparations.

A09a/B17 are complete. The crossover note derives a beta midpoint law and the exact interpolation from cubic short-window variance to the common long-window plateau. C038 proves that a fixed finite window and uniform speed ceiling cannot realize a positive shared coefficient over arbitrarily small masses. A09b supplies the subsequent direct checkerboard source passages and real-versus-complex two-component comparison.

A08/B16 are complete. The composition note proves that a nonnegative mass-only coefficient preserved by independent composition is constant on positive masses. A finite-rate, nonzero-speed reference makes the common Green–Kubo plateau positive. C035–C036 retain the preparation-class premise, and product-state closure accommodates more than two centre velocities. The original priority decision and its countertests follow.

The six-direction review promotes **A08** to the next main task: derive the mass-composition law for a nonnegative action coefficient and test its physical premises. Distinguish mass independence within one preparation class from independence of environment, and coefficient closure from closure of the composite trajectory law. The deliverable is a short proof note, five algebra checks and one sequential Luna-low prior-art audit. C033–C034 already belong to A06; new claims keep new IDs.

A09 combines the finite-speed crossover bound with the checkerboard source route. Verify real probability and complex-amplitude recurrences separately; compare stationary-increment and conditioned-midpoint observables. The B15 source companion records verified 1984 identifiers and remaining direct-passage obligations. The Compton-scale substitution uses an additional identification with $\hbar$; particle creation is a separate field-theory construction.

G01 now has a concrete finite-state model: the velocity-generator gap and its susceptibility product. Include a slow-mode observability countertest before seeking a lower gap estimate transferable to WP6. A03/A05/A06 close the chosen cut-consistency constructions; their observables remain distinct in the central selection problem. **A07** stays available as the bounded-force diagnostic. M03 retains its independent proof/build gates, and B11 is ready for its saved geometry audit. This reprioritization leaves the accepted claim set unchanged.

1.6 Starting results

A01 establishes a positive action plateau from finite reversible velocity memory: $H_* = 2m\langle v, (-Q)^{-1}v\rangle_\pi$, with the two-velocity value mu^2/λ. The paper and B09 audit separate established transport formulas from their action-valued consequences. **A02** adds physical-time covariance relaxation in an elastic bath, with plateau $(m+M)s^2/\nu$ set by reservoir parameters. The completed **A03** cut-point test proves that exact Gaussian restriction consistency fixes its fluctuation parameter, while the finite-defect scaling changes retained-node marginals. Each inserted node contributes mean kinetic action $\kappa/2$ at fixed preparation. **A05** adds an elastic physical-cut example and the sharp finite-speed midpoint bound $\kappa_{\mathrm{mid}} \leq m\Delta(u - |v|)^2$. Its convolution theorem identifies the need for memory in a stochastic finite-speed refinement model. **A06** now supplies a bridge retaining both position and velocity: the exact return law has a midpoint atom, consistent cut restrictions and vanishing polygon action error. **A07** replaces impulses by bounded acceleration and tests sharp return-duration and kinetic-action bounds at fixed endpoint velocities. The bath results remain diagnostics within the broader target. Necessity, convergence, universality and identification with $\hbar$ remain separate proof obligations. M06 and spectral examples support this central task.

The two-regulator calculation proves an exact finite action-defect limit. With $d = N - 1$ internal positions, $\kappa_N \to 0$ and $d\kappa_N \to \ell$, Gaussian polygonal bridges converge uniformly to the classical path while their excess action converges in L^2 to $\ell/2$. Classical Paths with a Finite Action Defect also supplies the partition determinant, a strict-speed oscillatory example, quadratic stationary-phase normalization and an explicit bare-coefficient map. B08 provides its per-result source and proof audit. M06 next tests force control and the relativistic kinetic action on the oscillatory sequence.

The current refinement experiment is an exact free-particle blocking map. Gaussian kernels compose while preserving their action parameter; endpoint bridges concentrate on the Newtonian straight path as that parameter tends to zero. Time Refinement and an Action Scale gives the proofs and identifies the open physical selection step. The source-to-model note connects this test to Plutarch's cone, Rivero's regulator proposal, Brouder's rooted-tree algebra and tangent-groupoid quantization.

Bibliography is an input to model design as well as a prior-art check. Every bounded reading should return a premise to test, a construction to implement, or a proof obligation to resolve. Read only the passages needed for the current calculation; use one small worker at a time, followed by coordinator review.

For $V = -Fy$, duration $T > 0$, perpendicular launch speed $v_0 > 0$ and mass $m > 0$, the chord–curve action difference is

$$\Delta S = \frac{F^2T^3}{24m} = \frac{F}{2v_0}\mathcal{A}_{\mathrm{lens}} = \frac{T\delta E}{12}, \qquad \delta E = \Delta K = -\Delta V.$$

The variation $\eta(t) = at(T - t)$ gives $\Delta S = ma^2T^3/6$. Thus positive action differences in this fixed-endpoint class have infimum zero. The chord is an off-shell comparison path; the parabola solves the constant-force equation. This identifies admissibility of neighbouring paths as a substantive choice for the proposed gap mechanism.

A second result concerns quantum measurement. For two known, equally likely two-arm pure states whose relative phase is $\Delta S/\hbar$, N independent copies and arbitrary joint binary measurements give the first-lobe threshold

$$d_{N,p} = 2\hbar \arccos\Big([4p(1-p)]^{1/(2N)}\Big), \qquad |\Delta S| \leq \pi\hbar, \quad \tfrac{1}{2} < p \leq 1.$$

Success probability at least p is achievable exactly when $|\Delta S| \geq d_{N,p}$. For fixed $\frac{1}{2} < p < 1$, this threshold scales as $N^{-1/2}$; perfect discrimination has threshold $\pi\hbar$ for every finite N. The calculation connects an action difference to a resource-dependent resolution.

The technical paper supplies the proofs, the Jacobi-operator calculation and the normalized free-kernel limit. The ledger records their evidence and review status.

1.7 Questions and mathematical objects

Target	Question	Starting point
Q0: classical existence	Which force regularity, initial data and collision conventions give unique global trajectories?	Constant force; next, central potentials
Q1: action-value gap	Is the infimum of positive matched-endpoint action differences positive on the chosen path class?	Infimum zero for the constant-force family
Q2: action scale	Which physical premises force a universal parameter with action units to be positive?	Reconstruction axiom comparison
Q3: operational bound	How do protocol, resources and error target determine action resolution?	Exact two-arm finite-copy threshold
Q4: spectral gap	Which self-adjoint operator has a separated ground sector, and how does its gap depend on parameters?	Dirichlet Hessian and quantum Hamiltonian
Q5: transfer across limits	Which estimates survive continuum and infinite-volume limits and interactions?	Companion PDE/field-theory comparisons

Use $h = 2\pi\hbar$ for Planck's constants, η for trajectory variations, J for a fluctuation operator and χ for a proposed additional field. Use matched endpoints when comparing actions under additions of $dG(q,t)/dt$.

1.8 Model sequence

1. **Constant force:** derive the area–action relation and study small variations.
2. **Harmonic oscillator:** compare the duration-dependent Hessian eigenvalues with the quantum Hamiltonian spacing. Track normalization and conjugate times.
3. **Free line, circle and box:** isolate confinement and boundary conditions, and calculate gap closure as the spatial domain grows.
4. **Central potentials and Kepler:** establish IVP assumptions, collision continuation and action-angle variables; then compare quantum spectra.
5. **Interacting and multiwell models:** identify a gap-producing mechanism and its dependence on coupling, tunnelling and limiting parameters.

1.9 Work packages

Package	Deliverable / acceptance gate	Dependencies
WP0: foundations	Restart instructions, ledger, tasks, source rules and LaTeX/PDF build	Complete
WP1: Newton	Edition/folio concordance, six-scholium corpus and dated evidence for the publication-history question	Source access

Package	Deliverable / acceptance gate	Dependencies
WP2: classical mechanics	Proofs for constant force, regular central forces and specified Kepler collision cases	Path class and topology
WP3: fluctuations and kernels	Jacobi operator/domain, conjugate times and normalized distributional limits	WP2
WP4: operational reconstruction	Measurement bounds and a dependency map of candidate quantum axioms	WP3; bibliography B01
WP5: gap laboratory	Operator domains, explicit bounds and parameter-dependent gap closure	Relevant WP2–4 results
WP6: companion limits	Relativistic, parabolic and quantum-field comparisons with a specified transferable estimate	WP5; Millennium definitions
WP7: papers and review	Readable proofs, checked citations, reproduced results and selected formal certificates	Accepted package result

Historical reading and mathematical calculations have separate task tracks. Delegate bounded tasks to Sol or Luna sequentially, with the coordinator waiting for each worker and reviewing its handoff before continuing. The finite-dimensional examples establish the definitions and estimates for the later field-theory comparison. A package finishes with a theorem, counterexample, conditional result or precisely stated open obligation.

1.10 Reconstruction branches

Branch A: consequences of quantum premises. Start with Hilbert states, Born probabilities and the phase $e^{iS/\hbar}$ for a specified $\hbar > 0$. Derive operational bounds, spectra and their dependence on model parameters.

Branch B: origin of quantum structure. Choose operational premises about composition, continuity, reversible transformations and distinguishability. Construct models satisfying subsets of these premises and identify the step that selects quantum structure. Then establish how action enters the dynamics and what forces its scale to be positive. Numerical calibration is a further empirical task.

Additional-field proposal. Specify the field space, coupling and measure for $S[q, \chi]$, then calculate the effective action after eliminating χ. The idea register sets out the candidate interpretations and their immediate tests.

1.11 Reading priorities

M05 and B07a have implemented the normalization audit of Rivero's 1998 proposal and the common-observable construction in Wilson–Kogut §12.2. The next source-to-model note turns Rindler's oscillation framework into M06's force-control test. B07b retains the precise tangent-groupoid quantization hypotheses and their classical-limit topology. This ordering joins source reading to a named calculation.

Start Q01 with Hardy's five axioms and Chiribella–D'Ariano–Perinotti's informational derivation. Use Masanes–Müller as a third comparison. For each, trace how its premises constrain state spaces, transformations and composition.

Branch A draws on Feynman's path-phase formulation, Wootters's finite-sample statistical distance and the quantum Chernoff asymptotic error bound. The B01 batch records verified metadata and selected reading coverage. Full reconstruction-proof reading belongs to Q01.

1.12 $1/c$, Navier–Stokes and Yang–Mills

Track $\hbar$, c^{-1}, viscosity ν, time and volume independently. The free relativistic particle retains the small-amplitude action family within a strict speed margin. The nonrelativistic quantum model retains $\hbar > 0$ at $c^{-1} = 0$. These examples separate the roles of the two parameters.

A bridge must specify its source and target spaces, operators, physical or auxiliary time, units, preserved estimate and limits. The useful comparison is how coercivity and control of fluctuations survive changes of scale. For Navier–Stokes, the target is evolution regularity under the official hypotheses. For Yang–Mills, it includes the physical Hilbert space and a vacuum spectral gap through the continuum and infinite-volume limits.

1.13 Scope and publication

The programme's reconstruction and gap-formation questions are open research targets. The present results concern the explicitly defined toy models; the Millennium problems supply comparison targets. Novelty and publication readiness will be assessed through literature comparison and specialist review.

Maintain human-readable proofs alongside calculations and formal artifacts. Record changes of model or target with their reasons. Submission, public posting, author correspondence and paid services require user direction. The user has authorized commits and pushes to the existing repository after each innovation or relevant status change (2026-09-07).